

Fig. 2: Actual-size PCB layout of fire alarm

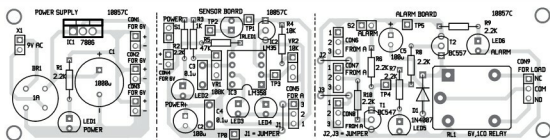


Fig. 3: Components layout for the PCB

TABLE I
TEST POINTS

Test point	Details
TP0	GND
TP1	6V
TP2	Up to 5V depending on IR radiation
TP3	450mV-550mV. Set using VR2
TP4	5V
TP5	6V

TABLE II
LED STATUS

LEDs	Status
LED1	'On' if 6V is available across CON1, CON2 and CON3
LED2	'On' if 6V is available in the sensor board
LED3	'On' if flame is detected
LED4	'On' if temperature is detected
LED5	'On' if neither temperature nor flame is detected
LED6	'On' if 6V is available in the alarm board

board can be placed at a safe and open area.

Initial setup

Once the sensor board (middle portion in Fig. 1) is ready and connected to 6V DC supply via switch S1, set VR2 to get around 450mV at TP3. (450mV means 45°C as set temperature, since LM35 outputs 10mV per 1°C temperature.) Now, using a hair dryer or some other small heat source, blow hot air on LM35. Once the temperature of LM35 crosses 45°C, LED4 glows. Stop the heat source. At pin 2 of IC3, you can set 450mV to 550mV for trigger temperature of 45°C to 55°C or any other temperature according to your requirement.

Next, light a candle and keep it about one metre away from IR LED1

sensor (flame sensor). Set VR1 such that LED3 goes off and glows again if the candle is moved little closer to the IR sensor.

Connect power supplies to sensor boards and the alarm board. Switch on S1 and S2. Test the relay operation by increasing the temperature and also by bringing the candle flame nearer to the IR sensor.

Construction and testing

An actual-size PCB layout of the fire alarm including power supply, sensor board and alarm board is shown in Fig. 2 and its components layout in Fig. 3. Cut the PCB along the dotted lines. Assemble the power supply, sensor board and alarm board separately. Enclose these in a suitable box. Connect the respective connectors

between power supply, sensor board and alarm board.

The circuit works off 230V AC, 50Hz. Alternatively, you can use a 6V battery. In this case, do not assemble power supply PCB and connect 6V battery directly at CON4.

After assembling the circuit, verify voltages as shown in Table I. Also refer Table II for the status of each LED in the circuit. Fix LEDs on the front panel and label their functions.

When you are using a single sensor board, connect CON1 to CON4 for power supply and CON5 to CON6 for signal sensing. In this case, CON2, CON3, CON7 and CON8 are unused.

For the arrangement using three sensor boards, you can assemble the sensor boards separately using the same sensor circuit and power supply taken from CON1, CON2 and CON3 respectively. **EFY**



Fayaz Hassan is a manager at Visakhapatnam Steel Plant, Andhra Pradesh. He has keen interest in microcontroller projects, mechatronics and robotics